

Hugo Leonardo

Full Stack Software Engineer | hugo.leonardo.jobs@gmail.com | linkedin.com/in/hugo-leonardo-p-a-495a40233

SUMMARY

8+ yrs experience in VoIP & cloud telephony building scalable SIP-based communication systems and full stack applications. Intermediate knowledge in blockchain and Web3 through learning projects.

SKILLS AND KNOWLEDGE

Languages:	JavaScript (ES6+), Python, HTML/CSS, Solidity (basic)
VoIP:	SIP, RTP, FreeSWITCH, WebRTC, SIP trunks, dialplans, extension routing, SIP trace analysis, softphone integration
Frameworks:	Node.js, React.js, Express.js, RabbitMQ, Bootstrap, jQuery, Passport.js
Web3/Blockchain:	Smart Contracts (concepts), Tokenization, DAOs (concepts), Web3 Integration, dApps (intermediate level)
Other:	Event-driven systems, Distributed architectures, RBAC, REST APIs, WebSockets, AWS (S3, SSM, SQS), Wireshark, Linux CLI, GitHub Actions CI/CD

PROFESSIONAL EXPERIENCE

Full Stack and VoIP Engineer — INOVA

2018 – Present

Specialized in FreeSWITCH, with expertise in SIP trunks, dial plans, and troubleshooting real-time voice services.

- Built a comprehensive fleet management system with WebRTC VoIP integration, real-time GPS tracking, and advanced troubleshooting tools, solving issues such as one-way audio, latency, and dropped calls.
- Developed full-stack architecture using RESTful APIs, WebSocket real-time communication, and a MySQL database storing GPS historical data. The platform includes geofencing, route optimization, and enterprise security using Passport.js with role-based access control (RBAC).
- Proven track record building reliable, mission-critical platforms (e.g., emergency medical and fleet management systems).
- Highly skilled in Node.js and event-driven architectures, utilizing WebSockets, RabbitMQ, and AWS.
- Architected SIP communication systems using FreeSWITCH. Managed the end-to-end telephony lifecycle, including the configuration of SIP trunks, complex dialplans, extension routing, and real-time user registrations.
- Built a fleet management VoIP solution. Supported about 200 vehicles.
- Solved problems in one-way audio, API, and dropped calls. Used Wireshark to troubleshoot issues.
- Designed and built feature to detect identity of driver when speed was exceeded.
- Built dashboard to show VoIP status.
- Developed the core voice communication engine for the SAMU Emergency Medical System, enabling high-reliability links between dispatch operators and field medical units.
- Built integrated SIP-based tools for fleet dispatch platforms, allowing seamless communication in Web command interfaces.
- Developed high-performance backend services using Node.js and Express, utilizing WebSockets for real-time data flow and RabbitMQ for event-driven messaging.
- Actively maintained system uptime by analyzing SIP logs and traces to diagnose and resolve complex signaling, routing, and registration issues within Linux environments.
- Implemented a production-ready DevOps pipeline with CI/CD, cloud deployment, and containerization using Docker, along with real-time analytics dashboards, automated maintenance scheduling, and a mobile-responsive Progressive Web App.
- Streamlined deployment workflows by implementing CI/CD pipelines via GitHub Actions and AWS.

LEARNING PROJECTS - BLOCKCHAIN/WEB3

Intermediate Level: Development of systems and prototypes for learning blockchain concepts, Web3, smart contracts and decentralized applications.

- Built asset tokenization system prototypes to understand concepts of fractional ownership and non-fungible tokens (NFTs).
- Developed study projects exploring smart contract architecture for simulated transactions, ownership distribution and staking mechanisms.
- Created experimental Web3 integrations with Node.js backends, including wallet connectivity (MetaMask), authentication flows and API design for dApps.
- Studied DAO governance models and decentralized platform architectures through reference implementations.
- Explored AI model integrations with blockchain data for market analysis and tokenized asset valuation.
- Investigated VR/AR immersion concepts applied to digital property experiences in metaverse environments.

EDUCATION AND TRAINING

Bachelor's Degree in Computer Science, UniCEUB

Certifications in React, JavaScript, AI and Blockchain